

Interim medical countermeasures network regional entities

Meeting report

Geneva, Switzerland
12–13 March 2024



World Health
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CORRIGENDA (18 November 2024)

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Replace [Page 4, Box 1, second bullet point, line 12-14] with the corrected sentence below

An estimated four to eight billion doses could be developed in the event of a pandemic. There are some treatments available but investment in this area is limited compared to vaccines.

Replace [Page 21, left column, line 14-16] with the corrected details below

WHO Health Emergencies Programme

WHO Regional Office for South-East Asia

New Delhi, India

These corrections have been incorporated into the electronic file.

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Abbreviations

ASEAN	Association of Southeast Asian Nations
ASPR	Administration for Strategic Preparedness and Response (United States of America)
BARDA	Biomedical Advanced Research and Development Authority (United States of America)
CARICOM	Caribbean Community
CARPHA	Caribbean Public Health Agency
CBRN	Chemical, biological, radiological and nuclear
CEPI	Coalition for Epidemic Preparedness Innovations
COVID-19	Coronavirus disease 2019
CSO	Civil society organization
Gulf CDC	Gulf Center for Disease Prevention and Control
INB	Intergovernmental Negotiating Body
MCM	Medical countermeasures
PAHO RF	Pan American Health Organization Revolving Funds
R&D	Research and development
RF	Revolving Funds
WHO	World Health Organization

Executive summary

Background

Interim medical countermeasures network (i-MCM-Net), a World Health Organization (WHO)-led interim coordination mechanism, addresses the urgent need for global preparedness against future pandemics, building on the lessons learned from the ongoing coronavirus disease (COVID-19) crisis. It aims to foster collaboration and ensure equitable access to medical countermeasures (MCM) through a network-of-networks approach, emphasizing inclusivity and partnership. In February 2024, WHO organized two virtual forums to update civil society organizations (CSOs) and industry stakeholders of the i-MCM-Net. Through discussions, the forums gathered insights on lessons from previous health crises to enhance preparedness to advance equitable access to MCM in the future. These discussions are summarized in Annex 3.

Subsequently, from 12 to 13 March 2024, WHO convened relevant entities of regional economic integration organizations (regional entities) in Geneva to:

- 1) further explore topics on advancing equitable access to MCM;
- 2) assess activities for emergency preparedness along the MCM value chain; and
- 3) outline potential collaboration between regional entities and WHO headquarters, regional and country offices.

As part of this meeting, five regional entities reflected on their mandates, initiatives and collaborations in relation to the end-to-end MCM value chain. Please refer to Annex 1 for list of participants and Annex 2 for the meeting agenda. The meeting of regional entities took place in the context of ongoing Member State discussions via the Intergovernmental Negotiating Body (INB) about access to MCM in health emergency preparedness and response. Whilst political negotiations are ongoing as part of the INB process, the focus of i-MCM-Net is to provide technical and operational insights.

At the request of Member States, WHO is collaborating with partners to assess the readiness of the MCM ecosystem for responding to pandemic influenza, novel coronavirus and pathogen X (a hypothetical, unknown pathogen). The aim is to identify gaps and suggest actionable steps to address them. Initial analyses show the need for enhanced capacities across the MCM value chain: in fast-tracked research and development (R&D), scalable manufacturing and seamless end-to-end supply chains for MCM. This includes transparent and accurate assessments of supply and demand, equitable allocation based on need, and building community trust and readiness for last mile delivery.

This report summarizes the discussions which took place as part of the meeting of regional entities.

Regional entities' roles in ensuring access to MCM

Regional entities occupy various roles across the value chain for MCM and work in different ways. Important topics highlighted across regions include localized manufacturing, pooled procurement and stockpiling. Below is provided an overview of the key functions of each entity involved in the discussions.

- The European Commission's Health Emergency Preparedness and Response Authority (HERA) works across the MCM value chain (supporting fast-tracked R&D, scalable manufacturing and end-to-end supply chains for MCM), facilitating R&D through the Delivering a Unified Research Alliance of Biomedical and public health Laboratories against Epidemics (DURABLE) consortium, while supporting adaptable and distributed manufacturing platforms through a network of European vaccine manufacturers (EU FAB), as well as a strengthened regulatory framework overseeing the set-up and scale-up of manufacturing platforms, establishing standards, policies and procurement mechanisms for transparent needs-based allocation and resilient logistics, amongst other activities.
- The Biomedical Advanced Research and Development Authority (BARDA) of the United States of America Department of Health and Human Services supports fast-tracked R&D through to licensure

and first procurement. It collaborates with other entities in the United States of America to inform and support processes related to downstream components of the MCM value chain.

- The Caribbean Community's (CARICOM) Caribbean Public Health Agency (CARPHA) supports end-to-end supply chains for MCM by engaging in various activities including maintaining a countermeasures list, conducting surveillance, shaping the market, aggregating demand, as well as supporting regulatory processes relating to procurement, needs-based allocation, distribution and quality assessment.
- The Pan-American Health Organization (PAHO) Revolving Funds (RF) supports end-to-end supply chains for MCM primarily through pooled procurement. Its functions include cost reduction, strengthening national capacities in demand estimation and supply chain management, ensuring continuous availability, maintaining quality standards, supporting adherence to PAHO/WHO regulatory mandates and providing financial options.
- The Gulf Center for Disease Prevention and Control (Gulf CDC) of the Gulf Cooperation Council's Gulf Health Council focuses on public health emergencies preparedness by offering risk assessments, scientific evaluations and technical guidance to inform Member State actions across the MCM value chain. It collaborates with the Gulf Health Council, which manages the Gulf Joint Procurement programme.

Considerations for geographically diversified production

Recent disparities in accessing MCM during public health emergencies have led to efforts among many Member States to establish localized capacity for production of medicines and health technologies. Expanding manufacturing capacities comes with additional costs and requires effective investment assessments, regulation and oversight to maintain product quality and meet global standards. It also requires consistent maintenance of operational facilities outside of emergency situations to ensure production readiness. Alternatives to localized R&D or manufacturing capacity can achieve similar outcomes and include technology transfers, cross-country collaborations to secure MCM doses, as well as reliance-based regulation.

Intelligence for demand planning and assessment of supply capabilities

Forecasting can inform decisions about procurement, stockpiling and manufacturing preparation. Forecasting requires a complete understanding of the supply chain including raw materials, facility capacities, specialized equipment and skilled personnel. Forecasting must also account for varying scale and likelihood of need under different scenarios for different MCM, including volume estimation and warm basing capacity. Prioritizing essential MCM requires decisions about discontinuing or deprioritizing other medications to support emergency response efforts. Dual emergency planning (e.g., pandemic and seasonal influenza) involves deciding what to procure with the available budget and what to develop during emergencies.

In practice, demand is affected by nuanced factors that may impact wastage and timing of need, such as varied dosing methods, fluctuating demand trajectories, preference and acceptability of vaccine type, vaccine efficacy and shelf life. Demand forecasting for MCM could benefit from more systematic methodologies and collaborative development of forecasting tools, potentially leveraging existing models. Supply assessment can be challenged by data quality; more so when assessing therapeutics and diagnostics compared to vaccines. The trust between industry and entities responsible for forecasting and procurement is a key factor in ensuring data reliability. Collaboration among regional entities can leverage relationships based on trust with manufacturers, thereby facilitating global access to reliable data.

Enablers of effective collaboration for end-to-end MCM systems

Effective collaboration on end-to-end MCM systems can be strengthened by enhancing transparency, facilitating coordination to mitigate resource competition and ensuring equitable allocation based on public health risks. Transparency aids in identifying pricing inequities, although its impact on medicine access outcomes remains nuanced. Transparency about the factors that inform price decisions may be valuable.

Strategic planning for dual-use products and raw materials, coupled with coordination efforts, can minimize competition for resources. Discussions around resource sharing centred on stockpile management and the feasibility of sharing and donating MCM, especially considering the importance of global and national health security. Regulatory hurdles need to be overcome, and agreements put in place, during inter-pandemic periods to facilitate effective resource sharing in emergencies.

Transparency in allocation is critical. Allocation varies across regions and requires different considerations for vaccines versus therapeutic products. Allocation of therapeutics from stockpiles requires ethical considerations for people who might become sick in the future and allocation efforts must remain free from political influence.

Supporting countries in ensuring equitable access to MCM

Discussions among regional entities considered how actions at both regional and global levels effectively support MCM access at country levels, recognizing disparities in regional support. While sub-regional entities can aid coordination, their effectiveness is limited by member resources. Global initiatives to address resource and capacity gaps are essential to assisting countries that lack comprehensive regional support.

1. Interim medical countermeasures network (i-MCM-Net) background

The impact of the COVID-19 pandemic continues to highlight collective vulnerabilities and the need to strengthen the international community's preparedness for responding to future pandemics. i-MCM-Net is a WHO-led interim coordination mechanism that aims to enhance collaboration for timely and equitable access to MCM against pandemic threats through a network-of-networks approach. i-MCM-Net aims to develop an ecosystem of partners to consider how we can ensure equitable access to quality, safe, effective and affordable MCM in the next pandemic. i-MCM-Net is currently directed towards assessing global readiness for equitable access to MCM in the event of a new pandemic threat. Drawing on lessons learned during the COVID-19 pandemic, i-MCM-Net aims to develop strategies during inter-pandemic periods that foster sustainability and equity in future response systems. It is important to note that certain principles and issues are within the jurisdiction of Member States, particularly those being discussed at the INB. i-MCM-Net focuses on providing technical and operational insights without pre-empting INB discussions.

The i-MCM-Net approach, led by WHO, is grounded in inclusivity, partnership, collaboration, listening and engagement. WHO has actively involved CSOs, industry and regional partners, committing to regular meetings for robust collaboration and consistent partner input. Since May 2023, there have been extensive efforts to engage stakeholders, including six consultations with Member States and five operational-level meetings involving 50 partners representing critical functions across the MCM process. Additionally, key stakeholders have been actively involved in considering how to jointly co-design and build a future-ready MCM system through engagement with civil society and participation in global meetings including the G20 in India, G7 in Japan and the Johannesburg process in Norway and South Africa. The MCM partners forum, which is continuously growing, encompasses representation from United Nations agencies, health organizations, civil society and non-governmental organizations, regional bodies (with whom WHO is aligning key strategies and portfolios), as well as industry and the private sector.

With a view to enhance coherence and foster synergies, WHO held two virtual forums in February 2024 for CSOs and industry stakeholders. The forums provided updates on the initiatives of i-MCM-Net and solicited input on learnings from past health emergencies about gaps and opportunities for enhancing preparedness towards ensuring equitable access to MCM in the future.

Following on these events, WHO hosted a meeting of regional entities in Geneva from 12 to 13 March 2024 to further discuss these topics, consider how regional entities are supporting emergency preparedness and response efforts across the MCM value chain, and identify how WHO can support such efforts. The meeting convened actors working to ensure equitable access to MCM from regional entities and WHO headquarters, regional and country offices.

1.1 Objectives

Recognizing the importance of regional capacities, the objective of this meeting was to provide updates on i-MCM-Net activities, to discuss preliminary findings and gaps emerging from the i-MCM-Net mapping report, to engage regional stakeholders in a collaborative discussion about challenges and to foster cross-regional dialogues.

Expected meeting outcomes included:

- establish cross-regional level communication channels and initiate an ongoing conversation among regional entities to improve collaboration for timely and fair access to MCM;
- contribute to an initial mapping of MCM activities along the value chain;

-
- provide input for a practical framework ensuring informed procedures for developing, producing, accessing, allocating and deploying MCM against pandemic threats.

2. The MCM landscape

In recent years, the number and intensity of epidemic, pandemic and other health emergency events, including those driven by climate change, have increased. Efforts to prepare MCM readiness and health systems for future emergencies are not just aimed at pandemics, but also at recurrent health crises and localized epidemics. Work undertaken by WHO Member States on governance for MCM financing hinges on negotiations within the INB and the Working group to the amendments if the International Health Regulations. These negotiations set fundamental rules for access, sharing and solidarity. The discussions of Member States also extends to leadership initiatives and accountability mechanisms. This includes working with the G20 Joint Finance and Health Taskforce, promoting sustainable financing for preparedness, as well as collaborating with multilateral development banks. These efforts aim to address the challenges of ensuring immediate emergency response financing, known as Day Zero financing.

2.1 MCM as part of a comprehensive health system response

The Health Emergency Preparedness and Response (HEPR) framework sets the background for efforts by WHO to enhance equitable access to MCM within health system contexts. (1) This framework encompasses collaborative surveillance, community protection, safe and scalable care, access to countermeasures and emergency coordination. Efforts to strengthen these systems focus on enhancing diagnostic and lab capacities, improving data quality, fostering collaborative surveillance approaches, co-creating public health measures for community protection, ensuring scalable and trustworthy care, fast-tracking research and development for countermeasures, establishing end-to-end health emergency supply chains, strengthening workforce capacity, in addition to improving global coordination for effective responses during health emergencies. Adopting a systems approach to these challenges requires aligning global and regional initiatives with national responses. Regional capacities are critical to consolidating and harmonizing these efforts.

2.2 Financing MCM

A joint analysis by WHO and the World Bank highlights a financing gap of US\$ 10.5 billion for pandemic preparedness, particularly in MCM. Despite various funding streams from bilateral donors, development banks and health financing initiatives, there remains a notable lack of private sector funding. Strengthening coordination is crucial to increase total funding, better coordinate investments and enhance efficiency. While significant funds exist for health emergencies, delays and coordination issues hinder their effectiveness.

Collaborative efforts to strengthen prevention and preparedness include initiatives led by WHO, the World Bank and the Pandemic Fund, as well as cross-country initiatives including the G20 Joint Finance and Health Task Force and the G7 Health Working Group. Access to Day Zero financing has been a particular priority for WHO and has attracted interest from various stakeholders. Day Zero financing could be made available through forward or advanced purchasing agreements of MCM, and at-risk financing via revolving funds or pooled funds. The capacity for Day Zero response relies not only on establishing these mechanisms but also on ensuring their ongoing use and sustainability. Inactive systems risk being overlooked, making it crucial to actively engage and refine emergency response systems during non-pandemic emergencies. Day Zero capacity can be achieved by consistently using funds and mechanisms to support localized outbreak and epidemic responses, thus keeping them primed for response to a global pandemic.

However, a key concern regarding MCM resourcing for pandemics is the potential impact on resource allocation for ongoing outbreaks and other needs. This issue is further elaborated in Box 6, which discusses considerations for stockpiling. To address these challenges, implementing an escalation system that triggers systematic actions for different scales of outbreaks—ranging from smaller local outbreaks to public health emergencies of international concern (PHEIC) or situations with pandemic potential—can be a valuable approach.

2.3 Mapping the MCM landscape



Upon request by Member States, an assessment is underway of MCM ecosystem readiness to respond to pandemic influenza, novel coronavirus and pathogen X (a hypothetical, unknown pathogen that could cause a future pandemic). The objectives of this activity are to identify key gaps and propose concrete actions needed to address them. Specific areas being examined for each disease area to determine readiness include:

- the R&D pipeline
- the production landscape and scalable manufacturing
- mechanisms for transparently and accurately assessing supply and demand
- approaches to ensuring equitable allocation based on need
- approaches to building trust among communities, and
- preparing communities to accept and receive MCM.

Leveraging their collective knowledge and network resources, the report on MCM ecosystem readiness is being co-developed by WHO in collaboration with key partners, such as FIND, Gavi, Unitaïd, the European Commission, the Coalition for Epidemic Preparedness Innovations (CEPI), the United Nations Children’s Fund (UNICEF), the PAHO revolving fund, as well as various CSOs and regional entities.

Initial analyses demonstrate that capacities are needed across several domains.

- **Fast-tracked R&D:** Essential work includes expediting R&D efforts across viral families, implementing operational R&D strategies for rapid clinical trials, and establishing regulatory systems to ensure swift quality assurance during emergencies.
- **Scalable manufacturing:** Despite the political will to regionalize manufacturing, sustainable infrastructure development is essential for long-term viability, particularly in peacetime to facilitate pandemic scale-up. Discussion is needed on investment strategies for regionalizing manufacturing capacity.
- **End-to-end supply chains:** There remains a lack of comprehensive insight into advanced purchasing agreements, with some countries providing information to WHO while others have not. Addressing this issue involves understanding procurement details, assessing export/import capabilities, as well as ensuring availability of necessary raw materials, along with resolving technical challenges through collaborative discussions.

Box 1 synthesizes specific findings for the three disease areas.

Box 1. i-MCM-Net assessment report: Disease-specific findings

Pandemic influenza

- Strategic collaborations for preparedness: There is a high level of confidence in the current state of pandemic influenza preparedness. Over the past decade, significant progress has been made including establishing a global research agenda and strategic measures for managing influenza-related events. The implementation of the Pandemic Influenza Preparedness (PIP) framework among Member States has also contributed to enhancing preparedness efforts.
- R&D: There has been considerable progress in influenza vaccine development and there is a surplus in R&D for vaccines. Several vaccines are already on the market and around 100 different vaccine candidates are currently in the pipeline. There are 30 influenza vaccine manufacturers globally, with seven manufacturers comprising approximately 85% of the global supply for seasonal influenza vaccines. An estimated four to eight billion doses could be developed in the event of a pandemic. There are some treatments available but investment in this area is limited compared to vaccines.
- Access and equity: High- and upper middle-income countries consume approximately 95% of seasonal influenza vaccines. Seasonal influenza vaccine utilization is an indicator of health system readiness to immunize the public against pandemic influenza, as evidenced during the 2009 H1N1 pandemic. The PIP Framework secures access to 10% of real-time global vaccine production during an influenza pandemic, as well as other pandemic products, with distribution based on public health risk and needs.

Novel coronavirus

- R&D: The COVID-19 pandemic triggered unprecedented MCM development. There has been significant investment in diagnostics and vaccines. Despite less investment in therapeutics, advances were made in the development of monoclonal antibodies during the COVID-19 pandemic.

Pathogen X

- Data to inform R&D: Developing MCM for unknown pathogens is challenging. Current efforts focus on creating scalable platforms for rapid response once a disease is identified. A strategy which emphasizes advancing surveillance and diagnostic tools to swiftly capture and provide essential information. Additionally, there is a concerted effort to build technology platforms and libraries. For example, the R&D Blueprint is investigating viral families to develop MCM with broad coverage and CEPI is creating a vaccine library. However, progress towards preparedness for Pathogen X is more evident in vaccine development than in treatments or diagnostics.

2.4 Political landscape for access to MCM

Access to MCM in health emergencies is under discussion at the INB meetings for a WHO instrument on pandemic prevention, preparedness and response. (2) INB talks on Articles 13, 19 and 20 frame the operational discussions which took place as part of this regional entities meeting.

2.4.1 Article 13 and 13bis

Article 13 of the draft negotiating text of the Pandemic Agreement puts forward the space for the creation of the Global Supply Chain and Logistics Network; 13bis addresses National procurement- and distribution-related provisions. These articles outline a system guided by the principles of equity, transparency, inclusivity, timeliness, fairness and consideration of public health needs to ensure robust and inclusive supply chain coordination. Considerable attention is being paid to the needs of developing countries, particularly those in fragile and humanitarian settings. Within Article 13bis, discussions revolve around regionalization, global access, technology transfer and reasonable stockpiling to facilitate equitable and timely access to MCM.

Key considerations include determining which products will be within the scope of the agreement and identifying which raw materials are critical for MCM production and which face risk of shortage. The articles cover procurement and transparency regarding both demand and production. Considerations explored include those relating to environments which may face greater difficulties when accessing products (e.g. humanitarian contexts), as well as considerations for product sharing, including mechanisms akin to those used in the PIP Framework, to ensure fairness and reciprocity among countries offering goods or services.

Issues relating to trade barriers and localization of manufacturing have been proposed for inclusion in Article 13, however there is a desire to retain the focus on only supply chain for tangible products.

2.4.2 Financing articles: Articles 19 and 20

There is a newly revised draft of the financing articles within the negotiating text for the pandemic agreement, specifically in Articles 19 and 20. The revised articles call to improve coordination among existing financing mechanisms through the establishment of a coordination mechanism that will enhance coherence and better leverage various financing sources. Harmonization and coordination of funding is emphasized in the Articles, alongside the need for pooled funding and mechanisms for information sharing, for example a dashboard of financing instruments.

3. Regional entities mandates, key initiatives and collaborations

Five regional entities reflected on their mandates, initiatives and collaborations in relation to the end-to-end MCM value chain. Across regions, entities occupy different components of the MCM value chain and provide different types of support. Regional entities with strong political and resource commitments were underscored as having greater independence in delivering large and diverse projects. All participating regional entities highlighted the diverse roles they have in pooled procurement and stockpiling.

3.1 The European Commission's Health Emergency Preparedness and Response Authority

The European Commission's Health Emergency Preparedness and Response Authority (HERA) was launched as a new European Commission Directorate-General in September 2021 in response to the COVID-19 pandemic with an aim to fill the void in institutional coordination for cross-border threats. Strong political support from Member States and a large budget to carry out its functions have supported HERA to deliver a wide portfolio of work across the end-to-end MCM value chain. HERA operates by anticipating health threats, with priorities including antimicrobial resistance, chemical, biological, radiological and nuclear (CBRN) threats, pathogens with pandemic potential, biosecurity and climate change.

HERA supports faster detection of health threats through improved data collection and surveillance. It also funds investments in innovative biologicals and technologies through initiatives like HERA Invest. HERA Invest is the first European health investment fund (€ 100 million) to promote advanced research and development of MCM and related technologies for tackling priority cross-border health threats including novel, re-emerging and expanding health threats resulting from climate change. HERA also funds innovation through initiatives including DURABLE (see Box 5). Additionally, HERA facilitates increased vaccine production through the EU FAB network, joint procurement agreements, as well as MCM stockpiling to address threats including CBRN, pathogens and antimicrobial resistance, amongst others. A Critical Medicines Alliance has been established to diversify MCM production in the region and enhance strategic collaboration in medicine procurement and distribution.

To support crisis phase response, HERA is streamlining the establishment of a Health Crisis Board. Operating at high government levels, this Board will serve to coordinate joint supply chains, collective procurement, emergency funding activation, in addition to utilization of EU-wide research and innovation plans, clinical trial networks and data-sharing platforms for a unified response to health crises.

Box 2. HERA support across the MCM value chain

HERA works across the three MCM priority areas of fast-tracked R&D, scalable manufacturing and supporting end-to-end supply chains as follows:

Fast-tracked R&D with pre-negotiated benefit sharing agreements including:

- coordinated research built on a shared global R&D agenda
- enabling environment for research and discovery through DURABLE
- standardized platforms for equitable and scalable clinical trials
- adapted regulatory and legal frameworks to enable timely trials, product review and approval.

Scalable manufacturing platforms and agreements for technology transfer including:

- adaptable manufacturing platforms
- distributed manufacturing capacity with pre-negotiated agreements to ensure equitable access
- ever-ready capability for rapid mobilization
- strengthened regulatory framework to oversee set-up and scale-up of manufacturing platforms.

Equitable and timely distribution of the right countermeasure, in the right volumes including:

- essential MCM and their associated standards, policies and enablers established for priority hazards through coordinated demand aggregation, coordinated supply and procurement, equitable and transparent needs-based allocation, as well as resilient logistics and distribution contracts.

Box 5. DURABLE as a knowledge resource to support HERA's decision-making

DURABLE, led by Institut Pasteur and the Erasmus Medical Centre, is a consortium comprising biomedical and public health laboratories in Europe. DURABLE focuses on experimental work in MCM development during preparedness and response phases. Utilizing metagenomic and genomics data, the network develops innovative modelling approaches based on laboratory-generated pathogen data. DURABLE supplements EU reference laboratories through interdisciplinary work packages and a pathogen-agnostic approach to prepare for and respond rapidly to Pathogen X. Institut Pasteur serves as the nucleus of a global network, with all institutions being European but having global reach, facilitating indirect global outreach. The network maintains active consultation and information exchange, enabling prompt signal detection, analysis and knowledge sharing.

3.2 Biomedical Advanced Research and Development Authority

Biomedical Advanced Research and Development Authority (BARDA) operates within the United States of America Administration for Strategic Preparedness, and Response (ASPR), focusing on the development and procurement of essential MCM relating to CBRN, pandemic influenza and emerging infectious diseases.

BARDA advances promising MCM developments from early clinical phases to licensure and initial procurement, particularly for products lacking a commercial market or facing hurdles in reaching the market. BARDA achieves this through public-private partnerships, prioritizing licensure to enable rapid MCM deployment for emergency response. However, despite the availability of a licensed product, maintaining the capacity for rapid scale-up remains challenging, underscoring the need for partnerships with product developers.

BARDA holds two key roles in emergency response efforts. First, when an event occurs and there is an insufficient supply of a licensed product, BARDA collaborates with partners to accelerate its manufacturing. Second, in instances where a new threat emerges and there are no existing MCM, BARDA is tasked with supporting its rapid development, as exemplified by the COVID-19 pandemic response. While BARDA's mandate does not encompass long-term stockpiling, it collaborates with the United States of America's Strategic National Stockpile and manages the supply chain for effective emergency response. BARDA partners with developers to license products for inclusion in the Strategic National Stockpile, with the expectation that they would be deployed in the event of an emergency.

BARDA's response to health emergencies differentiates between outbreaks of known infectious diseases and outbreaks of emerging infectious diseases (EIDs). Outbreaks of known infectious diseases are typically smaller scale than EIDs and comparatively easier to address in terms of funding and logistics. BARDA's role in responding to known infectious disease outbreaks primarily focuses on the late-stage development and approval stages of the medicine life cycle, with involvement in sustaining manufacturing and stockpiling. Other aspects, such as early MCM development, distribution and administration, are largely handled by partners within the United States of America's government or internationally.

In the context of an emerging pandemic, the lifecycle mirrors that of known infectious diseases but requires much more rapid implementation. In addition to the need for rapid development of MCM for EIDs, related challenges include global manufacturing capacity and sustained scalability, as well as the need for global collaboration.

BARDA noted how sustaining MCM post-licensure is a challenge when swiftly responding to an emerging pandemic. Sustaining readiness requires consistent infrastructure despite infrequent threats. Partnerships with manufacturers can aid sustainability, but their operations face uncertainties due to budget constraints

and unclear long-term commitments. BARDA highlighted how clarifying support duration and scale is vital for effective planning.

Potential solutions to ensure sustained scalability include leveraging adaptive technology and repurposing supply chains used for routine medical products to develop MCM needed during emergencies. However, BARDA described how this approach requires careful consideration. For example, repurposing production lines for vaccine development in emergencies may disrupt the production of other essential products. Moreover, ensuring a steady supply chain and access to raw materials is critical; any missing component could halt vaccine development.

BARDA highlighted how coordination among stakeholders is important to effectively navigate complexities and optimize the potential of adaptive technology for scaling up vaccine production in emergencies. While collaborative efforts are essential to response capabilities and industry assurance, they remain limited in number. Enhanced clarity on future procurement expectations can facilitate global preparedness efforts. The Ebola vaccine serves as a notable example of successful global collaboration, with multiple funders supporting various stages of research, development and procurement.

Several other areas of focus have been identified by BARDA to strengthen MCM for EIDs including exploring product attributes, such as developing cost-effective technologies or formulations requiring fewer doses and leveraging existing clinical trial infrastructure to better represent populations, while maintaining a warm manufacturing supply chain.

3.3 Caribbean Public Health Agency

As an institution of CARICOM, Caribbean Public Health Agency (CARPHA) was established in 2011 through the merger of five regional institutions to enable functional cooperation in health. CARPHA leads the response to public health emergencies and collaborates with the Caribbean Disaster and Emergency Management Agency to support the public health aspects of disaster response. CARPHA has supported efforts to respond to emerging health-related threats in the region, such as COVID-19, mpox and antimicrobial resistance, alongside recurring threats like vector-borne diseases, measles, norovirus and respiratory viruses.

CARPHA's activities are primarily funded by international development partners, with 31% of funding coming from Member State core funding. Its activities relating to MCM are situated downstream in the value chain, offering support for post-licensing and approval stages, including countermeasure list development, market shaping, procurement and distribution. Following licensing and approval, CARPHA engages in various activities including maintaining a countermeasures list, conducting surveillance and shaping the market, aggregating demand, procurement, needs-based allocation, distribution and quality assessment.

Noteworthy achievements include the development of an Integrated Surveillance Strategy, leadership in the regional health response to COVID-19, and the implementation of innovative programmes. For example, the Tourism and Health Programme, which creates cross-sector links so the tourism sector can report into the health system in real-time. Additionally, CARPHA fosters an enabling environment for effective partnerships.

CARPHA's role in supporting equitable access to MCM during emergencies is significant given the prominence of the tourism sector and extensive cross-Caribbean movement. Central to this effort is the Caribbean Regulatory System, which evaluates and recommends health care interventions within the region. The system plays a vital role in reviewing guidelines and supports capacity building, information dissemination, surveillance and market approvals. For example, during the COVID-19 pandemic, the

Caribbean Regulatory System developed recommendations and disseminated technical guidelines to Member States regarding the use of WHO-approved vaccines. All health care interventions introduced at the regional level undergo assessment by this system, ensuring safety and efficacy across the Caribbean community. CARPHA does not directly engage in pooled procurement but supports it within the region. For instance, the Organisation of Eastern Caribbean States, which carries out procurement, depends on CARPHA's joint regulatory reviews via the Caribbean Regulatory System for market approval. Additionally, they utilize CARPHA's services for post-market surveillance of approved products.

3.4 Pan American Health Organization (PAHO) Revolving Fund (RF)

The PAHO RF comprises a revolving fund and a strategic fund, collectively facilitating equitable access to timely, quality vaccines, essential medicines and public health supplies in the Americas (all countries except the United States of America and Canada). The PAHO RF primarily supports downstream components of the MCM value chain. Through pooled procurement, Member States benefit from affordable prices and coordinated demand planning. Guided by principles of solidarity, equity, quality, transparency and pan-Americanism, the RF aims to ensure MCM availability, affordability, scalability and sustainability. Its functions include cost reduction, strengthening national capacities in demand estimation and supply chain management. The PAHO RF ensures continuous availability, maintains quality standards, supports adherence to PAHO/WHO regulatory mandates, and provides financial options.

The PAHO RF provides distinct benefits for both Member States and manufacturers. It serves as a single point for consolidating demand, thus leveraging economies of scale. The RF incorporates a line of credit function, wherein a service fee of 4.25% is charged to cover administrative costs, as well as to finance a fund that offers Member States an interest-free loan, allowing them to pay for goods 60 days after delivery and invoice placement. Currently, US\$ 350 million is available in this fund for such financing purposes. Acting as a process facilitator, PAHO supports quality assurance and provides technical support for timely planning, financial projections, shipment monitoring and market signals. Additionally, it promotes solidarity by facilitating cooperation between countries.

The strengths of the PAHO RF lie in its advanced coordination efforts for demand planning and consolidation, supported by extensive collaboration with all Member States. The recent development of a digital platform enhances accessibility and further streamlines demand consolidation processes. Furthermore, significant progress has been made in allocation strategies, demonstrated by successful procurement experiences such as with the mpox vaccine during shortages, showcasing efficient resource allocation to Member States. PAHO is developing innovative mechanisms to facilitate agreements with manufacturers tailored for pandemic contexts, including volume guarantees and various procurement and financing instruments.

Additionally, PAHO RF is exploring stockpiling options. While the RF itself doesn't maintain a stockpile, PAHO does have an emergency stockpile, primarily donor-funded and deployed to countries as needed. Efforts are underway to establish a stockpile which would instead leverage domestic government resources, drawing on PAHO-facilitated procurement efforts totalling between US\$ 700 million to 900 million annually. The initiative aims to create a stockpile of key products, benefiting from economies of scale and addressing equity concerns. For example, recognizing that Caribbean Islands often face exorbitant transportation costs, sometimes paying up to 10 times the product's value, PAHO is investigating how stockpiling strategies can be tailored to benefit these Member States. Developing capacities for efficient distribution, storage and management of the stockpile is essential to maximize its effectiveness during emergencies.

The financial and legal functions within PAHO have facilitated the success of the RF. Established before WHO, PAHO's financial controls and legal framework have played a critical role in managing and overseeing

the operations of the RF. These functions ensure independence, transparency and effective governance, laying the foundation for the RF's efficient operation and success.

3.5 Gulf CDC of the Gulf Cooperation Council's Gulf Health Council

Gulf CDC was established in 2021 under the stewardship the Gulf Health Council. Its mandate is to address public health challenges in the Gulf region. Situated in Riyadh, Saudi Arabia, the strategic objectives of the Gulf CDC are to:

- support the development of collaborative public health programmes, policies and practices
- strengthen regional public health training and capability building
- facilitate surveillance and research, and
- enhance preparedness, early detection and rapid response to health hazards in the Gulf region.

Gulf CDC's MCM efforts primarily address public health emergencies preparedness, in addition to research and training programmes. Built on close collaboration, Gulf CDC has also established a regional event detection and risk management cycle, focusing primarily on outbreaks and pandemics, with a mandate covering all health hazards, including environmental health. Based on a recent country consultation, Gulf CDC is currently developing both a preparedness plan and a response plan. Four key objectives of these efforts include:

- strengthening regional training and knowledge sharing
- developing regional public health guidelines (also relevant to MCM)
- rapid and coordination emergency response, and
- unlocking the value of public health assets, data and infrastructure.

Gulf CDC provides guidance and assessments to inform national and local implementation of action and activities, including rapid risk assessments to help countries understand public health events and provide recommendations for immediate actions. Gulf CDC also provides technical guidance and scientific assessments to inform efforts towards manufacturing localization and stockpiling for key medical products, which are priority areas for the six Gulf Cooperation Council member states - Bahrain, Kuwait, Oman, Qatar, Saudi Arabia and United Arab Emirates.

Gulf CDC operates under the Gulf Health Council, receiving operational support such as corporate communications finance, and procurement from the council. The Gulf Health Council, established in 1976, serves to promote and enhance the health sector across its member states, which include the aforementioned six member states of the Gulf Cooperation Council and Yemen.

Through the Gulf Joint Procurement programme, one of its earliest implemented initiatives, the Gulf Health Council has established mechanisms for procuring various medical supplies, medicines and vaccines, emphasizing safety, quality standardization, providing alternative sources, market intelligence, exchange of surplus products between the countries and promoting localized production. Thus far, 20 000 items (over 20 tenders) valuing US\$ 3 billion have been procured from over 600 manufacturers through this mechanism.

The advantages of a joint procurement programme include simplified regulations, pre-set procurement cycles, shared workload among countries, enhanced transparency, the ability to conduct emergency tenders and a high level of automation integrated into the process. The selection criteria for pharmaceuticals include that the drugs must be registered through a central mechanism in place through the Gulf Health Council or in two of the reference countries, one of which should be Saudi Arabia. Additionally, the manufacturers themselves should be centrally registered or registered in one of the reference countries. For medical devices, the criteria differ slightly but align with similar principles. Medical devices must be evaluated and accepted in at least two countries, and manufacturers must be centrally

registered or registered in the Saudi Arabia. Following initial screening, compliance with procurement terms and conditions, sample quality, conformity to specifications, commitment to supply terms, after-sales service for each country and price fairness are considered.

3.6 Summary of MCM capabilities across entities

Table 1 provides an overview of existing MCM capabilities or MCM capabilities currently being expanded by regional entities. The table explores capabilities through HEPR access to MCM categories.

Table 1. MCM capabilities being expanded by regional entities (existing and underway).

Entity name	Span of countries	R&D	Manufacturing	MCM Lists & Market Shaping	Demand	Procurement ^a	HEPR Capabilities					
							Allocation	Stockpiling	Distribution	Integrated Delivery	Regulatory Harmoniz. / Strengthening	Regulation
Africa Centres for Disease Control and Prevention (ACDC), including the African Pooled Procurement Mechanism (APPM)	Africa		x	x	x	x (APPM)		x	x	x		
Caribbean Public Health Agency (CARPHA) and Organization of Eastern Caribbean States (OECS)	Caribbean			x	x	x (OECS)	x		x	x	x	x
EU Health Emergency Preparedness and Response Authority (HERA)	EU	x	x	x	x	x	x	x	x			
Gulf Center for Disease Control (CDC) and Gulf Cooperation Council (GCC) Joint Procurement Program for Pharmaceuticals and Medical Supplies	Gulf				x	x (GCC)	x		x	x	x	
PAHO Regional Revolving Funds (RRF)	LAC		x	x	x	x	x	x	x			
US Biomedical Advanced Research and Development Authority (BARDA), US Administration for Strategic Preparedness & Response (ASPR)	USA	x	x	x	x	x	x	x (ASPR)	x			

^a R&D and Manufacturing reflect activities to incentivize and support these, not necessarily conducting R&D or manufacturing.

Data notes: Africa CDC was not able to attend the meeting, but the information is included given their ongoing engagement in i-MCM-Net; For CARPHA and Gulf CDC, the associated procurement entities are included even though they did not attend the meeting.

A range of capabilities are employed by regional entities, illustrating the importance of collaboration and coordination across regional, national and global entities. Collaboration and coordination are critical to ensure sufficient deployment of the capabilities needed to ensure access to MCM, as well as to optimize resource allocation and promote equitable access to MCM. Within coordination, there are important differentiated and complementary roles.

- National level: Countries identify local needs, conduct research focused on national health priorities, implement policies and deliver MCM directly to communities. They provide critical on-the-ground insights and data, which inform broader strategies. Countries have the legal authority for MCM regulation.
- Regional level: Regional entities can coordinate efforts among neighbouring countries, facilitating the sharing of resources, information and best practices. They can address cross-border health issues, enable regional supply chains more efficiently and use collective bargaining to improve access.
- Global level: Global organizations set international standards, provide guidelines and mobilize global resources and funding. They support research and development, ensure equitable distribution of MCM and coordinate international responses.

By working together, a robust and resilient system will drive timely and fair access to MCM.

4. Priority areas for ensuring equitable access to MCM

4.1 The role of regulation in enabling sustainable and geographically diversified production

Strengthening local production of medicines and other health technologies is recognized by many Member States as a key strategy to increase timely access to medicines. Over 100 Member States supported World Health Assembly Resolution WHA74.6 on Strengthening local production of medicines and other health technologies to improve access. (3)

Meeting discussions with regional entities also highlighted Member State interest in localized manufacturing capacities. In several cases, prioritizing localized capacity emerged from inequities experienced accessing MCM in recent health emergencies. For example, in the Caribbean, discussions on vaccine production addressed limited access and inequities experienced during the COVID-19 pandemic and mpox outbreak. Discussions underscored challenges faced by middle-income countries in accessing resources if income levels and national capacities are too high to attract MCM via donor funds and yet not enough to attract pharmaceutical investment. PAHO RF emphasized that given regional interest, options for supporting manufacturing in the region are being explored. In the Gulf region, Gulf Cooperation Council is promoting localized production and highlighted that numerous initiatives and activities are underway at the national level to localize production, establish sufficient agreements with MCM producers and consider stockpiling measures.

Discussions emphasized that the expansion of manufacturing capacity requires various complementary capacities.

- **Regulatory capacity** plays a crucial role in localizing manufacturing processes to ensure reliable pharmacovigilance outcomes. Regulatory systems tasked with monitoring manufacturing quality wield considerable influence over the efficacy and safety of medicines. The agency overseeing pharmacovigilance outputs must demonstrate sufficient reliability and capability to monitor manufacturers in adherence to established standards. Regulatory systems can be established, implemented and scaled during inter-pandemic periods, so they are primed to effectively respond to crises.
- **Sustainable manufacturing capacity** should take precedence over simply expanding it. During the peak of the COVID-19 pandemic, many Member States expressed ambitions to become leading producers of vaccines, highlighting a widespread desire for self-sufficiency. However, such aspirations may prove challenging in the long term given issues of cost and profit associated with establishing new facilities; the potential for technology transfers and cross-country agreements to achieve similar objectives; the need to keep manufacturing facilities warm and operational to ensure cost effectiveness and rapid scale up. In Europe, the HERA EU FAB initiative provides an interesting case study for how such objectives are being achieved (See Box 3).
- **Capacity to meet WHO benchmarking standards** is essential yet, in countries where there is significant state or regional level fragmentation, only certain medications meet the WHO benchmarking standards.
- **Capacity to assess substandard and falsified medicines** is critical and regulatory oversight is needed given the heightened risk of investment in these products during emergencies. The February 2024 i-MCM-Net CSO Engagement Forum also highlighted this challenge, noting that quality assurance and product testing is in some cases covered by CSOs rather than national authorities. When responding to pandemics, relying solely on existing tools for quality assurance may not suffice; the introduction of new, adaptable tools capable of facilitating faster responses becomes imperative, as does the need to validate these novel tools.

Box 3. EU FAB contributes to ever-ready capability for rapid mobilization

Established by HERA, the EU FAB network comprises vaccine producers across the EU with agile and sufficient manufacturing capacities to quickly produce essential vaccines in the event of a health emergency. The objectives of this network are to:

- close the gap between manufacturing and scaling up of vaccine production
- ensure the capacity of the industry to produce life-saving medicines
- reserve manufacturing capacities, and
- obtain a priority right for manufacturing of vaccines in case of a future public health emergency.

EU FAB comprises private vaccine production facilities reserving capacity exclusively for MCM during emergencies, capable of producing mRNA, vector-based, and protein-based vaccines. Member entities maintain readiness to swiftly commence manufacturing upon the European Commission's request. The network ensures a total capacity of 325 million doses per year during crises, aiming to produce 50% within the initial six months of activation. Producers are from Belgium, Ireland, the Netherlands (Kingdom of the) and Spain. With an annual budget of € 160 million, EU FAB ensures adaptable manufacturing capacities for various vaccine types and facilitates rapid activation.

Global collaboration on regulatory capacity is crucial for enabling rapid access to MCM in an emergency. Significant efficiencies can be found in a system where countries can trust and rely on regulatory decisions and oversight from other agencies.

Considerations for supporting this goal emerged during the regional entities meeting. Firstly, WHO supports the possibility of reliance-based regulation through monitoring efforts using the Global Benchmarking Tool for evaluation of national regulatory systems and Prequalification and Emergency Use Listing processes. These processes were particularly valuable in supporting regulatory alignment for COVID-19 vaccines. Evaluations were carried out by WHO and through transparent data-sharing with regulatory authorities, reliance principles were promoted through direct pathways as well as through regional networks to enable in-country approval for use & post-listing monitoring of COVID-19 vaccines.

Additionally, WHO Prequalification and Emergency Use Listing criteria can be better leveraged by empowering regional entities to assume more regulatory responsibilities. Strengthening existing capacity could facilitate this expansion.

4.2 Intelligence gathering and “how to”

Access to data and intelligence gathering informs preparedness across the MCM value chain. Regional entities discussed approaches to quantifying MCM needs and demand, assessing supply capabilities (including available manufacturing facilities and production volumes), projecting future requirements and developing demand scenarios. Key reflections on demand and supply assessments are presented below.

Assessing demand

- **Timing of demand:** Demand assessment, for vaccination especially, requires consideration of both the total number of doses required and the timeline for when specific quantities need to be delivered to achieve immediate and longer-term goals.
- **Factors affecting real world demand:** Whilst initial estimates of COVID-19 demand used calculations of one shot per arm, practical application revealed nuanced variations that can affect demand calculations in the emergency context, different to what is forecast during the MCM development phase. Demand trajectory fluctuated over time, with initial surges followed by declines, leading to potential vaccine wastage. Global demand forecasting for COVAX had to

consider countries' willingness to accept different vaccines, considerations about vaccine efficacy and shelf life.

- **Concurrent emergencies:** Planning for a dual emergency situation (e.g., pandemic and seasonal influenza) can be achieved to some extent by determining what can be stockpiled in advance, based on available budget, versus what needs to be developed during an emergency.
- **Comprehensive supply chain understanding:** The entirety of the supply chain, including raw materials and facility capacities, is crucial to forecasting efforts. Prioritizing essential MCM may require discontinuing or deprioritizing other medications that share the supply chain to facilitate emergency response.
- **Specialized manufacturing:** MCM manufacturing relies on specialized equipment and skilled personnel, with significant lead times and costs. Exploring global antigen manufacturing capacity and assessing scalability and adaptability could provide valuable insights into enhancing preparedness and response capabilities.
- **Demand estimations and policy alignment:** Discussions explored the basis of the forecast—whether epidemiological (e.g., targeting high-risk groups), economic (e.g., addressing international entry/movement requirements) or societal (e.g., facilitating the easing of public health and social measures). The importance of linking demand forecasts to policy setting and support was emphasized. Additionally, it was noted that basing demand forecasts on the minimum requirements for high-priority groups might limit available financing and production capacity to the allocated amount. This approach risks creating inequitable access if the industry uses these forecasts to determine production capacity during supply shortages.
- **Systematized methodology for demand forecasting:** Systematic methodology, tools for estimation and aggregation, and clear mathematical modelling approaches for such assessments are lacking, particularly in ways that are understandable to decision-makers. There would be value in collaboratively developing such tools, potentially leveraging tools used for routine modelling and assessment. Reference groups may also be valuable for ensuring consistency in demand modelling across regions. WHO may have a role in supporting such collaborations and standardizing models used for decision-making.
 - **Equitable allocation:** To reduce the risk of limiting supply, demand forecasting models, calculations should assume that everyone who needs the vaccine will receive it.
 - **Divergence in estimation approaches:** Demand estimation varies by purpose, such as targeting high-risk groups or easing public health measures. Challenges remain in balancing technical accuracy with political and public health objectives.
- **Scope of diseases being forecasted:** Estimating the needs for Pathogen X should not overshadow demand assessments for MCM for controlled diseases that may resurge, such as dengue. Even with available diagnostics, planning for potential future demand remains essential.
- **Data and information needs of manufacturers:** Considering the cost of warm basing (i.e., maintaining manufacturing capacity) versus scaling up production, for diseases like mpox, the primary expense lies in warm basing to sustain capacity. While for COVID-19, with an inability to stockpile, focus shifts to scaling up capacity for production. Understanding manufacturer requirements involves determining the balance between warm basing and production scaling, which varies depending on the disease.

Assessing supply

- **Data quality challenges:** Evaluating supply involves grappling with data quality issues, particularly in therapeutics and diagnostics compared to vaccines. Industry's willingness to share data may be influenced by how data will be used. For vaccines, data use is more evident (e.g. data is needed to understand capacities for rapid scaling) but for other products it is less clear.

- **Relationships and trust:** Gathering data from manufacturers poses challenges, with trust being a crucial determinant of data confidence. Collaboration among regional entities can leverage informal trust relationships with manufacturers, enabling global access to reliable data.
- **Capacity for distribution and deployment:** Challenges in deploying COVID-19 antivirals stem from logistical complexities, including establishing clinical guidelines for various scenarios and practicalities of implementation. Capacity for planning and rollout readiness is essential, emphasizing the need to proactively address logistical considerations before MCM arrival.

4.3 Collaboration considerations for global, regional and national end-to-end MCM systems

Effective collaboration on end-to-end MCM systems can be enhanced by improved transparency across multiple factors, improving coordination to prevent resource competition and facilitating equitable allocation based on public health risks and needs.

Transparency

- **Transparency can enhance awareness and identify inequities in pricing, but considerations on this topic are nuanced and there is mixed evidence that greater transparency can support achievement of desired outcomes in medicine access.** On the contrary, Member States may opt to keep prices confidential, even at the expense of efficiency, to secure access.
- **Transparency about the factors which inform price decisions can be more informative than prices themselves.** Discussion highlighted how price alone doesn't necessarily convey meaningful information; disclosing conditions for lower prices is more insightful. There is a need to understand what prices represent and volume data alongside price data can be more informative. Regional entities shared a variety of experiences in relation to advocating for greater price transparency. For example, joint procurement processes used in the Gulf region use differential pricing among members, but the exact prices paid are not known among other members.
- **WHO has undertaken efforts to enhance transparency** by collecting data on the cost of seasonal influenza vaccines, reported by countries themselves, and anonymized by gross domestic product (GDP) grouping and region. This annual dataset is considered a global public good, facilitating non-Gavi countries' self-reporting and mutual benefit through shared information.

Resource sharing

Strategic planning for products and raw materials with dual uses, along with coordinating efforts with partners, is crucial to minimizing resource competition. Discussions around resource sharing focused on stockpile management (see Box 6 for further considerations relating to stockpiling approaches) and the feasibility of sharing and donating MCM. These discussions also rely on effectively balancing global versus national health security.

- **Extracting resources from national stockpiles is highly challenging.** Stockpiles are created by entities to fulfil their mandate for their own population. For example, though BARDA may have surplus in their stockpile, donating from that supply can limit its ability to fulfil its mandate if there is future need for those products in the United States of America.
- **Regulatory, legal, liability and practical barriers hinder the feasibility of donations.** Challenges surrounding liability and regulations hinder recipient countries from accepting MCM donations, as observed during the mpox outbreak. Despite willingness from European and American authorities to donate MCM, the absence of established procedures limited immediate access. Negotiating and implementing procedures including process and conditions for donations would be a valuable step towards enabling access in emergencies. This is further elaborated in Box 4.

- **Stock rotation should be explored as an approach to resource sharing.** Supplies with shorter expiry dates could be transferred from countries with non-urgent needs to those with urgent requirements. Subsequently, countries could reciprocate by returning supplies with longer expiry dates in the future, facilitating efficient medical resource utilization.
- **A central global stockpile may be an effective mechanism for resource sharing.** For example, through the World Food Programme a centrally held stockpile of food and relief items is available for international use. This allows for swift turnover of both funds and stock through simplified legal agreements, facilitating efficient and rapid deployment during emergencies. However, there would be challenges in WHO enabling a similar function for MCM, as WHO primarily offers technical assistance. Once MCM arrives at a country's border, it becomes the country's responsibility to ensure its safe distribution to where it's needed. Safety and pharmacovigilance also fall within the remit of individual countries.
- **Stockpiles can be leveraged as a pull-through mechanism to influence markets,** particularly for vaccines in the trial stage. Gavi is leading such discussions. There is ongoing concern within CEPI about investing in R&D for new areas while considering the implications for stockpiles and procurement strategies.

Box 4. The need for effective distribution pathways

Equitable access to MCM hinges on securing adequate resources and using them effectively. This involves navigating complex distribution processes, including addressing liability issues and implementing regulatory procedures for inter-country donations.

However, the COVID-19 pandemic highlighted challenges: delays and coordination issues hindered the impact despite substantial funding; Late stock delivery, sometimes after demand had decreased, required significant reprogramming of financing.

In the Caribbean, despite successful efforts to secure vaccines and effective pathways for distribution within countries, challenges persisted in physically delivering COVID-19 vaccines to countries due to transportation constraints exacerbated by reduced flights and economic limitations during the COVID-19 pandemic. To enhance pandemic preparedness and response, CARPHA included a component addressing transport challenges in its pandemic fund application.

Box 6: Approaches to stockpiling

Regional entities highlighted a range of approaches in how stockpiling is managed.

For example, HERA assists members in stockpiling MCM to combat various threats by providing grants. Currently, the focus is on CBRN threats, with 11 Member States actively stockpiling. Each Member State adopts its own approach to stock management. PAHO maintains a donor-funded stockpile for regional use. The need for regional capacity for stockpiling emerged during COVID-19 pandemic with the notable need for a stockpile of syringes.

How to approach stockpiling requires consideration of the risks and benefits and weighing these against risk appetite. For example, larger stock can be maintained but may be more costly and require expired products be discarded. Alternatively, virtual stockpiles can be maintained with manufacturers, but this may come at a premium cost. Further development in this area is warranted.

Allocation

Regional entities take various allocation approaches, considering that vaccine allocation presents different logistical, equity and ethical considerations than therapeutics allocation.

- **Allocation of therapeutics from stockpiles requires ethical considerations for people who might become sick in the future.** Using national stockpiles during surplus periods could result in shortages if therapeutics are needed later. Ethicists from the United States of America National Institute for Health stressed the importance of equitable management, that access to treatment should not depend on the timing of illness. During the COVID-19 pandemic, conveying this message to leaders and health care providers proved challenging. There is a misconception that surplus availability resolves allocation issues, disregarding the need for ethical and health-based resource allocation processes. Strategies should be developed to prepare health systems for effectively managing this dilemma in future health crises.
- **Allocation efforts must remain free from political influence.** To achieve this, allocation frameworks should incorporate safeguards enabling decision-makers to prioritize public health independently. Implementing independent advisory groups could help. Allocation should translate policy decisions into technically sound implementation, closely tied to scientific guidance.
- **Various entities employed different approaches to vaccine allocation.** In the United States of America, vaccines were distributed among 64 jurisdictions based on a federal formula considering population and product availability. The COVID-19 response faced criticism for prioritizing equality over equity in vaccine allocation by allocating doses equally by population share. Some approaches used multiple criteria in their allocation strategy, for example by considering additional factors like HIV prevalence to enhance equity or including functional factors like vaccine benefits. Overall, however, data collection inconsistencies and uncertainties limited what factors could be included in COVID-19 vaccine allocation decision-making. Increasing consideration of upstream variables such as availability and shelf life complicates allocation.
- **Transparency in allocation is critical.** Despite COVAX's implementation by WHO and Member States, concerns emerged about its opacity, raising doubts about fairness and methodology. In the Americas, PAHO communicated allocation criteria to Member States early, promoting transparency and trust.

4.4 Country-level support

Discussions among regional entities considered how actions at both regional and global levels effectively support individual countries.

- **Supporting countries outside regional entities involves addressing resource and capacity disparities.** Many countries lack the resources to manage MCM access independently, relying heavily on organizations like WHO. Regions encompassing varying income levels and country sizes have trouble in finding a common mechanism for coordination. In the Western Pacific Region, access to nationally allocated resources is particularly limited, hindering efforts to expand stockpiles and enhance preparedness. Both the Regional Office for the Western Pacific and the Regional Office for South-East Asia are exploring a common platform, potentially supported by the Association of Southeast Asian Nations (ASEAN), yet institutionalizing these efforts poses challenges despite expressed support from some countries, for example Thailand's fund allocation for health emergencies.
- **Sub-regional entities can aid in coordination, yet their effectiveness is constrained by member resources.** For instance, the Eurasian Economic Union faces uncertainties regarding middle-income countries' self-sufficiency in handling health emergencies. In cases where regional support is lacking, global initiatives like COVAX and efforts led by UNICEF or WHO can aid countries.

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- **Regional coordination mechanisms can offer valuable resources to participating countries, even if utilization varies by country.** PAHO's pooled procurement initiatives exemplify this, leveraging favourable conditions among participants to secure optimal agreements. Although initially involving just 10 out of 42 Member States, it now includes 41. However, not all Member States exclusively procure through PAHO; some Member States with local manufacturing capacity, like Argentina and Brazil, opt for national procurement in some cases. Sharing experiences across regions can help establish similar mechanisms elsewhere, contingent upon political will.

5. Next steps

The meeting marked a first step towards enhancing regional collaboration for equitable access to MCM. Discussions emphasized potential areas for further cooperation. WHO aims to subsequently convene a series of engagements and technical meetings with regional entities, as well as a global partners meeting on MCM. This gathering will focus on developing a comprehensive global workplan and establishing mechanisms for monitoring progress.

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Annex 1. List of participants

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Annex 2. i-MCM-Net regional entities coordination meeting agenda

Date and Time: Tuesday - Wednesday 12 - 13 March 2024 CET

Organizer: World Health Organization (WHO)

Venue: In person at the Geneva Press Club (Le Club Suisse de la presse (CSP))
Chemin de l'Impératrice 18, 1292 Pregny-Chambésy, Geneva Switzerland

Duration: 2 days

Participants:

- Regional entities and Specialized Agencies, intergovernmental global and regional organizations – Africa CDC, ASEAN, BARDA, CARPHA, Gulf CDC, and HERA.
- WHO relevant departments including those working on the i-MCM-Net.

Moderation by: Dr Ioana Ghiga, Technical Officer, Epidemic and Pandemic Preparedness and Prevention, WHO Health Emergencies Programme.

Rapporteur: Ms Krista Kruja

Agenda

Day 1. Tuesday, 12 March 2024

9:00 – 9.15	Opening and welcome • Dr. Maria D. Van Kerkhove, Director (a.i), Department of Epidemic and Pandemic Preparedness and Prevention, WHO Health Emergencies Programme
9:15 – 9:45	Introduction • Participants • Agenda and objectives of the meeting • Housekeeping
9:45 – 10:30	Session 1 – Presentations on WHO MCM state of play • WHO Health Emergency preparedness, prevention and response architecture ○ Mr Scott Pendergast, Director, Strategy, Programmes and Partnerships, WHO Health Emergencies Programme • WHO interim coordination mechanism for MCM ○ Tim Nguyen, Head of Unit for High Impact Events, Epidemic and Pandemic Preparedness and Prevention, WHO Health Emergencies Programme
10:30 – 10:40	Keynote remarks ○ Dr Mike Ryan, Executive Director, WHO Health Emergencies Programme
10:40 – 11:00	Health break
11:00 – 11:30	Continue session 1 – Presentations WHO MCM state of play

	• Pandemic financing options ○ Mr Scott Pendergast, Director, Strategy, Programmes and Partnerships, WHO Health Emergencies Programme • Reflection from INB 8 on article 13 “Global Supply Chain Network,” ○ Mr Paul Molinaro, Head of Unit for Operations Support and Logistics, WHO Health Emergencies Programme
11:30 – 12:15	Session 2 – Regional entities mandates, key initiatives and collaborations • HERA - Scope, mandate, activities along MCM end-to-end value chain 25min presentation + 20min Q&A ○ Mr Boriša Falatar, Programme Coordinator, Health Emergency Preparedness and Response (HERA), European Commission
12:15 – 13:30	Lunch break
13:30 – 14:15	• BARDA - Scope, mandate, activities along MCM end-to-end value chain 25min presentation + 20min Q&A ○ Dr Robert Johnson, Director of Medical Countermeasures Program, Biomedical Advanced Research and Development Authority (BARDA) ○ Ms Jessica Swenson, Interdisciplinary Scientist, Antimicrobials Branch, Biomedical Advanced Research and Development Authority (BARDA)
14:15 – 15:00	• Africa CDC - Scope, mandate, activities along MCM end-to-end value chain 20 min presentation + 10 min Q&A ○ (TBC) Dr Abebe Genu, Acting Coordinator, Partnerships for African Vaccine Manufacturing (PAVM), Africa Centres for Disease Control and Prevention (Africa CDC)
15:00 – 15:15	Health break
15:15 – 16:00	• Gulf CDC - Scope, mandate, activities along MCM end-to-end value chain 20 min presentation + 10 min Q&A ○ Dr Pasi Penttinen, Chief Executive Officer, Gulf Center for Disease Prevention and Control (Gulf CDC)
16:00 – 16:45	• ASEAN - Scope, mandate, activities along MCM end-to-end value chain 20 min presentation + 10 min Q&A ○ Dr Ferdinal Fernando, Assistant Director & Head of Health Division, Association of Southeast Asian Nations (ASEAN)
16:45 – 17:30	• CARPHA - Scope, mandate, activities along MCM end-to-end value chain 20 min presentation + 10 min Q&A ○ Dr Horace Cox, Assistant Director of Surveillance, Prevention and Control, and Head of Vector-Borne Diseases, Caribbean Public Health Agency (CARPHA)
17:30 – 19:00	Network reception and adjourn day 1

Day 2. Wednesday, 13 March 2024

9:00 – 9.15 **Recap of day 1**

- Mr John Lee, Technical Officer, High Impact Events Unit, Epidemic and Pandemic Preparedness and Prevention, WHO Health Emergencies Programme

9:15 – 10:45 **Session 3: Discussion on sustainable and geographical diversified production**

- Discussion among partners about ongoing initiatives, investment and collaboration in sustainable and geographical diversified production
- Moderation by Dr Rogério Gaspar, Director of the Department of Regulation and Prequalification, Access to Medicines and Health Products Division, WHO

10:45 – 11:15 **Health break**

11:15 – 12:15 **Session 4: Intelligence gathering and “how to”**

- Identifying the methods and sources of safe, effective and quality assured products including the necessary raw materials and potential surge capacities as well as developing and maintaining a tool for this purpose
- Moderation by Ms Shanelle Hall, (the Yellow House)

12:15 – 13:30 **Lunch break**

13:30 – 15:00 **Session 5: Collaboration considerations for global, regional and national end-to-end MCM systems**

- Promoting transparency in cost, pricing and other relevant data on products, including raw materials, across the value chain;
- Promoting and coordinating to avoid competition for resources among international procuring entities, including regional organizations and/or other mechanisms;
- Identifying national, regional, and/or international stockpiles of various products
- Facilitating the equitable allocation of products during emergencies based on public health risks and needs
- Moderation by Dr Ioana Ghiga, Technical Officer, Epidemic and Pandemic Preparedness and Prevention, WHO Health Emergencies Programme

15:00 – 15:15 **Health break**

15:15 – 16:00 Session 2 continued – Regional entities mandates, key initiatives and collaborations

- **PAHO RF** - Scope, mandate, activities along MCM end-to-end value chain

25min presentation + 20min Q&A

- Mr Santiago Cornejo, Executive Manager, Revolving Funds (RF), Pan American Health Organization (PAHO)

16:00 – 16:45 Session 6: Country support tools, guidance and training

- Assisting countries in meeting the requirements for the effective utilization of pandemic products, as needed and requested
- Moderation by Mr Benedict Millinchip, Senior Lead, Global Health Security (Vaccines), WHO

16-45 – 17:00 Next steps and closure of meeting

Annex 3. CSOs and industry forums

In February 2024, the WHO i-MCM-net mechanism held two virtual forums for representatives of CSOs (22 February) and industry actors (23 February) to engage in cross-sectoral dialogues with the intention of considering the challenges and potential key areas of action for strengthening pandemic preparedness and response.

The forum for CSOs was moderated by Ms Sarah Hess, Technical Officer, Epidemic and Pandemic Preparedness and Prevention, WHO Emergencies Programme. A total of 254 participants from 63 countries registered for the event and 90 attended on the day. Participants represented primarily CSOs and non-governmental organizations, as well as United Nations agencies, health organizations, government and academia. The forum for industry actors was moderated by Dr Ioana Ghiga, Technical Officer, Epidemic and Pandemic Preparedness and Prevention, WHO Emergencies Programme. A total of 97 participants from 20 countries registered for the event and 63 attended on the day. Over half of participants were from the private sector and about one quarter from United Nations agencies.

Summaries of the presentations and discussions which took place at each event are included below.

i-MCM-Net CSO engagement forum note for record

Dr Mike Ryan, Executive Director of the WHO Health Emergencies Programme opened the event by highlighting the vital role of CSOs in representing communities and emphasized their importance in discussions about building a resilient ecosystem for future health crises. Cautioning against the danger of “preparing for the last pandemic, not the next one”, he called on participants to draw on lessons from the past, including COVID-19 pandemic, whilst also anticipating different health challenges. In a world of coalitions and alliances, we need to work collectively. Through meetings like this, stakeholders can establish a shared understanding of the current landscape and its gaps, and jointly agree a vision for the future. Noting that “no single actor possesses all the solutions, means or science to make pandemic preparedness a reality”, Dr Ryan called on participants to bring together efforts on behalf of those we serve: our populations and our communities.

Dr Maria Van Kerkhove, acting Director for the Epidemic and Pandemic Preparedness Department, also underscored WHO's commitment to meaningful dialogue and collaboration, with an aim to actively integrate the insights and expertise of CSOs into decision-making processes. Together with CSOs, WHO aims to provide people with accurate information, necessary tools and access to life-saving interventions. While scientific advancements are vital, they are only valuable for communities if interventions are made accessible, affordable and are taken up.

Mr Tim Nguyen, the Head of Unit for High Impact Events, Epidemic and Pandemic Preparedness and Prevention Department in the WHO Health Emergencies Programme reported on the development of the i-MCM-Net Mapping Report, which assesses the readiness of the MCM ecosystem to respond to pandemic influenza, novel coronavirus and pathogen X. The report, co-developed by WHO and key partners, reflects each partner's knowledge, input and network resources, as well as describing five key areas to determine readiness. These are: the research and development pipeline; the production landscape and scalable manufacturing; mechanisms for transparently and accurately assessing supply and demand; approaches to ensuring equitable allocation based on need; and approaches to building trust among communities and preparing communities to accept and receive MCM.

Mr Scott Pendergast, the Director of Strategy, Programmes and Partnerships in the WHO Health Emergencies Programme shared updates on the Pandemic Fund and WHO's efforts to assess financing needs for preparedness and rapid response to pandemics. Key priorities include improving coordination for financing, including among multilateral development banks, and increasing investment in surveillance, laboratory systems and the health emergency workforce. Options are also being explored for ensuring access to Day Zero financing, in which resources would be made immediately available to partners to initiate activities on the ground during a crisis.

Meeting discussions subsequently centred around presentations delivered by Dr Rosemary Mburu, Executive Director of WACI Health and Co-Lead of the Inclusive Global Health Institutions Project, Dr Richard Neci Cizungu, Executive Director of the Ecumenical Pharmaceutical Network, and Ms Eloise Todd, Executive Director and Co-Founder Pandemic Action Network on lessons learned from CSOs that can be applied to enhance pandemic preparedness and response. Speakers drew from their real-life experiences supporting communities through past health emergencies. Presentations and participant discussions which followed highlighted several lessons.

- Communities should be considered throughout all stages of pandemic preparedness and response, ensuring minimum standards are met to facilitate meaningful participation of civil society in decision-making processes (see **Box A1**). A human rights-based approach to ensuring access to MCM should centre the lived experiences of affected people (see **Box A2**).
- During the COVID-19 pandemic and other health crises, CSOs have been instrumental in addressing gaps and inequities (see **Box A3**). Communities play a crucial role in reaching marginalized populations overlooked by health and social systems, and their services should be leveraged to access hard-to-reach groups. However, government action is still necessary; systemic changes must address root causes of inequalities and gaps in services (see **Box A4**).
- Implementing links between stakeholders during interpandemic stages is essential for swift response during health emergencies. Strategic collaborations between CSOs and other stakeholders should be established beforehand to leverage the role CSOs can have in promoting MCM uptake in communities (see **Box A5**). Additionally, early communication of MCM development to communities (before their deployment) can foster the trust needed for effective uptake.

Box A1. Minimum standards for CSO engagement

During the COVID-19 pandemic, global health institutions engaged civil society as per their existing mechanisms and practices. However, it is not sufficient to rely on only a few global health institutions having strong engagement mechanisms. Some minimum standards to facilitate meaningful participation of CSOs in decision-making processes are as follows:

- CSOs should have a seat and voting rights (not observer-only seat);
- CSO engagement needs to be funded to facilitate participation and meet equity objectives;
- sufficient lead time is needed when sharing materials to enable CSO representatives to consult with their communities and bring their views to the table;
- input from CSOs should be treated with integrity and included in the final product.

Box A2. Drawing on a human rights-based approach to ensure communities are central to health emergency responses is essential for fostering trust

The HIV response has long been guided by a human rights framework, which centres on the recognition of the agency held by members of marginalized communities and promotes their visibility. In the HIV response, CSOs and people living with HIV have been critical to advancing access to human rights, and thus spurring increased action to develop and ensure access to essential MCM.

Anchored in international human rights law, the right to participation extends across all health domains. Whilst the commitment to inclusivity and participation has been institutionalized in the HIV response, with civil society representation formally integrated into governance structures, community participation was not fully leveraged during the COVID-19 pandemic. Countries that demonstrated resilience during the pandemic had the unique quality of bringing together various efforts (e.g. including national government, international donor programmes, and other actors) to support robust community-led responses. However, in many instances, communities were engaged only reactively, such as to address low vaccine uptake. Waiting until vaccines are available in communities to initiate communications can undermine community trust and uptake. In cases where there was early engagement of community leaders, such as religious leaders, in disseminating information about vaccine development and efficacy using accessible language, this proved effective in promoting understanding and acceptance within communities.

Box A3. The role of CSOs in filling gaps

During emergencies like the COVID-19 pandemic, the surge in demand for health products can lead to concerns over product quality, particularly with the proliferation of falsified or low-quality items. Expedited distribution efforts often overlook quality assessments; for example, in the rush to distribute gloves as personal protective equipment during the pandemic, thorough evaluation of thickness or quality was skipped by many service providers. To address this gap in quality assessment, the Ecumenical Pharmaceutical Network implemented the minilab, a simple tool comprising two briefcases and trained health care workers, to monitor product quality. Beyond testing quality, the minilab raised awareness, particularly in rural areas where access limitations and pricing pressures are particularly strong forces that can compromise quality oversight. This service did not only benefit church-based health services, but quality monitoring results were shared with the broader community and national health systems to maximize benefits. Additionally, the increased awareness achieved through this process facilitated informed decision-making and helped combat the spread of falsified or substandard medical products.

Box A4. Systems change to address MCM availability challenges

Low and middle-income countries faced barriers to accessing vaccines once they were developed during the pandemic, even with the financial resources needed to purchase them, since doses had already been ordered by and promised to high-income countries. One solution to this problem included examining the potential for dose donations. However, what made the biggest difference were agreements to implement technology buyouts which led to expanded vaccine production capacity. The success of this approach underscores the importance of prioritizing structural solutions to inequity-related challenges.

Box A5. Procurement solutions for smaller health service delivery actors

A functional procurement mechanism is essential for managing supply chain disruptions during a pandemic. Church-based health service providers faced challenges with procurement; the Ecumenical Pharmaceutical Network procurement initiative encountered difficulties in confirming tenders due to fluctuating availability, prices, and competition with larger buyers. However, given that charity and private sector services provide 40% of health care in some countries, collaboration among these actors could make pooled procurement initiatives feasible in the future. WHO can support such efforts by forging the connections needed among various faith-based organizations to achieve this goal.

i-MCM-Net industry forum note for record

The meeting was opened by Dr. Maria D. Van Kerkhove, acting Director for the Department of Epidemic and Pandemic Preparedness, and introductory remarks were made by Dr Mike Ryan, Executive Director of the WHO Health Emergencies Programme. Dr. Ryan emphasized the significance of Member State negotiations surrounding a Pandemic Accord, noting it as a collective effort over two years to establish mechanisms that can prevent and prepare for future pandemics. The existing MCM ecosystem holds extensive capabilities in innovation, logistics and research. By convening stakeholders, including the dynamic private sector, i-MCM-Net aims to jointly assess the strengths, gaps and roles of various actors in the MCM landscape to define shared objectives and approaches to strengthen the existing system. Dr. Ryan emphasized that our shared goal, whether in the public or private sector, “is to serve our clients by ensuring their protection, and our clients are our communities”. This ongoing dialogue marks a step towards that goal.

Mr Tim Nguyen, the Head of Unit for High Impact Events, Epidemic and Pandemic Preparedness and Prevention Department in the WHO Health Emergencies Programme reported on the development of the i-MCM-Net Mapping Report, which assesses the readiness of the MCM ecosystem to respond to pandemic influenza, novel coronavirus and pathogen X. The report, co-developed by WHO and key partners, reflects each partner’s knowledge, input and network resources and describes five key areas to determine readiness. These are: the research and development pipeline; the production landscape and scalable manufacturing; mechanisms for transparently and accurately assessing supply and demand; approaches to ensuring equitable allocation based on need; and approaches to building trust among communities and preparing communities to accept and receive MCM.

Mr Scott Pendergast, the Director of Strategy, Programmes and Partnerships in the WHO Health Emergencies Programme shared updates on the Pandemic Fund and WHO’s efforts to assess financing needs for preparedness and rapid response to pandemics. Key priorities in this space include improving coordination for financing (including among multilateral development banks), increasing investment in surveillance, laboratory systems and health workforce and ensuring partners on the ground have rapid access to resources during a crisis via Day Zero financing. Sustained and blended financing involving both public and private sectors is essential. Meeting participants were encouraged to reflect on what are the financing gaps and what collaborative approaches can be taken to address them.

Meeting discussions subsequently centred around presentations delivered by Dr John C. Pottage, Lead Scientific Consultant at Intrepid Alliance, Dr Rajinder Suri, CEO of Developing Countries Vaccine Manufacturers Network International, Dr Dennis Hruby, Chief Science Officer for SIGA Technologies and two presenters from AdvaMed - Dr Ashley Miller, Executive Vice President of Global Strategy and Analysis, AdvaMed and Zack Rothstein, Executive Director of AdvaMedDx.

Speakers drew from real experiences relating to MCM development and provision to highlight key challenges and strategies which can be used to address them as follows:

- **Data and evidence are needed to accurately assess and consequently make decisions on supply and demand, scalable manufacturing, medicine distribution and equitable allocation of MCM.** Equitable MCM allocation based on need relies on monitoring, data, and analysis to understand health needs and availability of MCM solutions both during and between health crises. During the COVID-19 pandemic, there were frequent calls by medicine purchasers for increased production of MCM, but manufacturers’ efforts were hindered by a lack of information about the quantities of such medicines that were needed and by when. Such analysis is essential during and between pandemics to inform policy, funding and efforts to address potential gaps (**Box A6 and Box A7**).

- **Equitable vaccine R&D and manufacturing financing is needed.** Despite developing countries producing 60% of COVID-19 vaccines, they received only 5% of the US\$ 5.6 billion research funding made available. Local and regional vaccine manufacturing is critical for addressing large-scale public health challenges. Manufacturing capacity across regions needs to be expanded in inter-pandemic periods and funding during pandemics should be directed to manufacturers across country income levels (**Box A8**). Vaccine R&D should also be de-risked. Recognizing that most products will fail during the development process, the most efficient approach to delivering a successful vaccine relies on progressing many diverse candidates through human trials.
- **Streamlining processes throughout the medicine lifecycle can overcome regulatory barriers to approval, manufacturing and distribution,** particularly arising from complexities due to variation in regulations among countries and regions, which can hinder access and exacerbate global drug shortages. Steps which can be taken during inter-pandemic stages towards this goal include:
 - R&D: Cross-country use of globally recognized standards and agreement on standardized approaches to carrying out clinical trials for diverse pathogen families.
 - Manufacturing and distribution: Establishment of universal packaging and labelling, and approaches to universal standards for prescription information to speed up production, packaging and distribution in an emergency.
 - Country-level approval: Establishment of regulatory pathways for known pathogen variants, agreed approaches to reliance models for regulation and permanent implementation of mechanisms such as the regulatory stability mechanism, which facilitated over 5000 approvals during the COVID-19 pandemic.

Different types of MCM come with unique challenges. Medical technologies face pronounced supply chain challenges due to potential raw material overlap with other industries, leading to competition. Regulation is lacking altogether in many countries for medical technologies, and deploying medical technologies requires health systems to develop capacity for personnel to operate and maintain specialized equipment. See **Box A9** for elaboration of obstacles faced with therapeutics.

Across MCM, WHO can play a critical role in facilitating the connections and mechanisms needed to overcome shared challenges including:

- facilitate coordination between Member States to streamline manufacturing processes;
- assess the utilization of regulatory flexibility measures during the COVID-19 pandemic and develop guidelines for implementing similar flexibilities in future pandemics; and
- map and assess the best available data systems in Member States to assist in evaluating demand surges during health emergencies.

Box A6: Landscape mapping to identify gaps in research and development

A consortium of seven pharmaceutical biotechnology companies, dedicated to expediting the antiviral pipeline for pandemic preparedness, have been mapping the current landscape to identify what compounds are in development and could potentially be deployed rapidly to respond in the event of new health threats, and where there are gaps. Across all development stages, various products are in development and approved for coronaviruses and influenza, but few address other viral families. Reports of these findings are made available online at the Intrepid Alliance website (<https://www.intrepidalliance.org>), providing information about the name of the drugs in development, the mechanism of action and sponsor. Reports will be updated quarterly, enabling ongoing identification of priority needs. This information can be used to develop shared assessments of priority targets and R&D strategies, such as target product profiles, to streamline efforts and guide clinical trials, highlighting the importance of establishing uniform or global target profiles for informing drug development in anticipation of future pandemics.

Box A7. Information sharing to enhance efficiency of MCM distribution during a pandemic

During the COVID-19 pandemic medicines were being stopped at borders, expiring in government stockpiles or otherwise failing to reach patients that need them. This raises questions such as: What data or information would give governments the confidence to avoid stockpiling of medicines that are needed to treat patients elsewhere, and what can be done to strengthen the mechanisms that can support these decisions? How can manufacturers more quickly, and with greater predictability, move medicines to where they are needed?

In early 2020, the International Generic and Biosimilar Association and WHO set up a dialogue to address these challenges and discuss collaboration about medicines shortages, emerging patient needs and the manufacturing response from the generic and biosimilar industry. Steps were taken towards:

- information sharing about demand surges, supply chain disruptions and border closures;
- engagement with WHO and other stakeholders to find solutions to supply medicines for patients; and
- engagement with Member States to increase the output of prioritized medicines.

Box A8. What is needed to equitably expand vaccine manufacturing capacities in low and middle-income countries

Using technology platforms and end-to-end capabilities which exist among manufacturers in Southeast Asia and the Western Pacific regions can significantly enhance preparedness for future pandemics. The three major enablers of R&D and manufacturing for vaccines in low and middle-income countries are the establishment of R&D hubs, building sustainability through investment and funding and collaborations for scaling up and scaling out. Regulatory flexibilities and expanding industry engagement in vaccine R&D is crucial to enhance manufacturing capacity and ensure affordable, equitable access in low and middle-income countries. To achieve this, the private sector needs:

- sustained funding from early development stages for R&D;
- joint and expedited clinical trials;
- regulatory harmonization and reliance mechanisms for faster registration in all countries;
- pro-active mapping of post-licensure use and demand, drawing on data which maps real needs and available resources;
- removing barriers to effective scale up by improving local supply chains and delivery systems; and
- facilitating collaboration between manufacturers and public authorities.

Box A9. Access challenges for TPOXX during the global mpox outbreak, 2022

Experiences with TPOXX, a therapeutic which was used to treat mpox, underscore the challenges in accessing therapeutics across the end-to-end value chain, requiring close collaboration among governments and agencies worldwide to address them.

- **Proactive need for R&D:** TPOXX's 15-year development process highlights the importance of proactive planning in medicine development. Regulatory agencies also demanded evidence of real-world effectiveness post-approval, extending trial durations.
 - Critical attributes for treating viruses with pandemic potential include treatment availability, long-term stability, formulations accessible to diverse populations and indications allowing immediate administration upon exposure, particularly in cases with extended disease incubation periods.
- **Supply chain management:** While initially developed for smallpox, TPOXX proved effective during the mpox outbreak, emphasizing the importance of maintaining sufficient drug stocks for real-time responses. Ensuring access to the raw materials needed for drug development was a critical first step.

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- **Slow distribution:** Lengthy import permits, regulatory approvals and complex shipping logistics led to delays in getting TPOXX to patients, resulting in some individuals either recovering or experiencing worsened consequences by the time the drug arrived.
 - **Trust to enable uptake:** Widespread awareness of TPOXX's development as it was ongoing was critical for enabling its uptake during mpox. Communications about the development and safety of therapeutics even in R&D stages can mitigate the need for extensive communication campaigns to build trust in that product during an emergency.

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